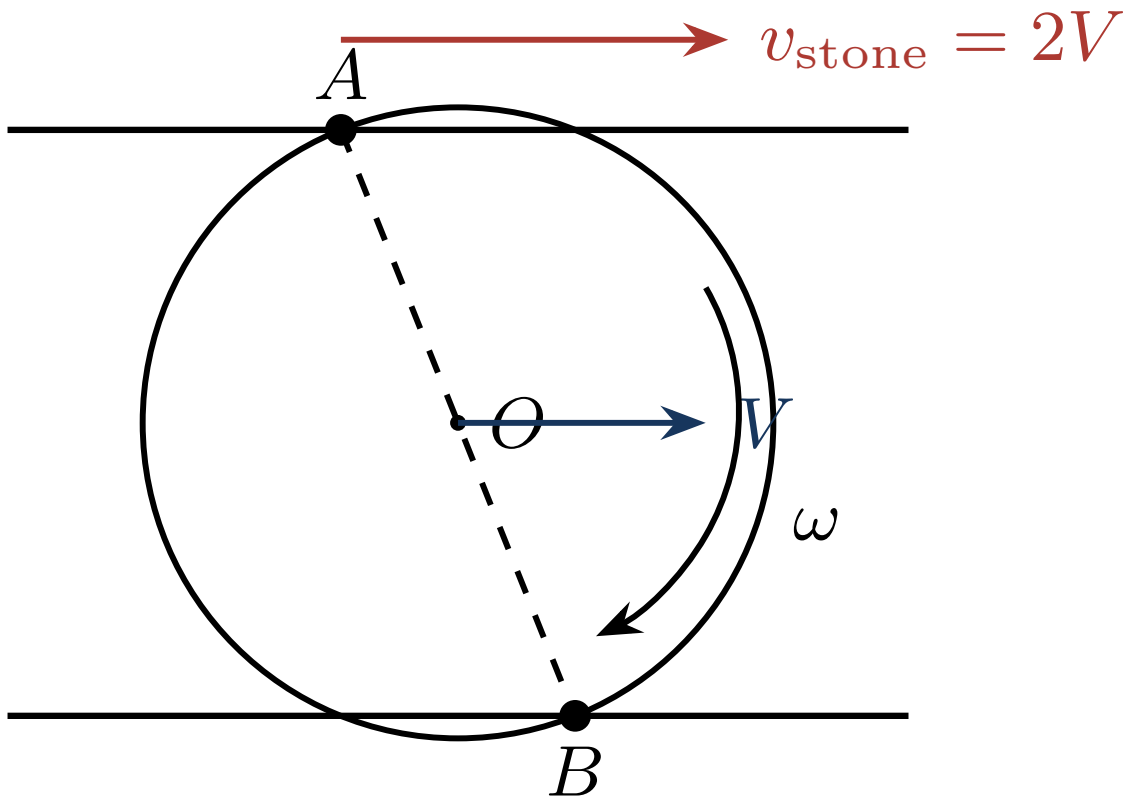

下接触：$V - \omega r\cos\theta = 0$；上接触：$V + \omega r\cos\theta = 2V$。